COMPUTER VISION

LAB SHEET :03

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```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# EXPERIMENT 3
# Spatial Filtering Techniques

# Load the image
img = cv2.imread("image.jpg")

# Check whether image was loaded
if img is None:
    print("ERROR: Image not found!")
    print("Make sure image.jpg is in the same folder as exp3.py")
    exit()

print("Image loaded successfully!")

# Convert BGR to RGB
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# LOW-PASS FILTERS

# 1. Gaussian Filter
gaussian = cv2.GaussianBlur(gray, (5, 5), 0)

# 2. Median Filter
median = cv2.medianBlur(gray, 5)

# 3. Average / Mean Filter
average = cv2.blur(gray, (5, 5))

# HIGH-PASS FILTERS

# 4. Laplacian Filter
laplacian = cv2.Laplacian(gray, cv2.CV_64F)
laplacian = cv2.convertScaleAbs(laplacian)

# 5. Sobel X
sobel_x = cv2.Sobel(
    gray,
    cv2.CV_64F,
    1,
    0,
    ksize=3
)
```

```
sobel_x = cv2.convertScaleAbs(sobel_x)

# 6. Sobel Y
sobel_y = cv2.Sobel(
    gray,
    cv2.CV_64F,
    0,
    1,
    ksize=3
)

sobel_y = cv2.convertScaleAbs(sobel_y)

# DISPLAY RESULTS

plt.figure(figsize=(14, 8))

# Original Image
plt.subplot(2, 4, 1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

# Grayscale
plt.subplot(2, 4, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale")
```

```
plt.axis("off")

# Gaussian
plt.subplot(2, 4, 3)
plt.imshow(gaussian, cmap="gray")
plt.title("Gaussian Filter")
plt.axis("off")

# Median
plt.subplot(2, 4, 4)
plt.imshow(median, cmap="gray")
plt.title("Median Filter")
plt.axis("off")

# Average
plt.subplot(2, 4, 5)
plt.imshow(average, cmap="gray")
plt.title("Average Filter")
plt.axis("off")

# Laplacian
plt.subplot(2, 4, 6)
plt.imshow(laplacian, cmap="gray")
plt.title("Laplacian Filter")
plt.axis("off")

# Sobel X
plt.subplot(2, 4, 7)
```

```
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X")
plt.axis("off")

# Sobel Y
plt.subplot(2, 4, 8)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y")
plt.axis("off")

plt.tight_layout()
plt.show()

# QUESTIONS AND ANSWERS

# 1. What is spatial filtering? How is it used in digital image processing?
# Answer: Spatial filtering is a technique in which the pixel value of an image is modified
# based on the values of its neighboring pixels.
# It is used for image smoothing, noise reduction, edge enhancement, sharpening,
# and improving important image features.
# Spatial filtering is commonly performed using convolution kernels.

# 2. Differentiate between Low-Pass Filters and High-Pass Filters with suitable examples.
# Answer: Low-pass filters reduce high-frequency components and smooth the image.
# They are mainly used for noise reduction and removing small details.
# Examples of low-pass filters are Average Filter, Gaussian Filter, and Median Filter.
# High-pass filters emphasize high-frequency components such as edges and fine details.
# Examples include Laplacian and Sobel filters.
```

3. Compare Average Filter, Gaussian Filter, and Median Filter based on their working principles and applications.

Answer: The Average Filter replaces each pixel with the average of its neighboring pixels.

It provides simple image smoothing but may blur edges.

The Gaussian Filter uses a Gaussian-weighted kernel and provides smoother and more natural

noise reduction while preserving important structures better than the Average Filter.

The Median Filter replaces each pixel with the median value of its neighborhood.

It is especially useful for removing salt-and-pepper noise while preserving edges.

4. Why is the Median Filter particularly effective for removing salt-and-pepper noise?

Answer: Salt-and-pepper noise appears as random black and white pixels in an image.

The Median Filter replaces each pixel with the median value of neighboring pixels.

Since extreme noisy values do not strongly affect the median, the noise can be removed effectively.

At the same time, important edges are preserved better than with simple averaging.

5. Explain the role of convolution kernels in spatial filtering.

Answer: A convolution kernel is a small matrix of numerical values used to process

the pixels in an image.

The kernel is moved across the image and its values are multiplied with corresponding

pixel values and then combined to produce a new pixel value.

Different kernels are designed for different purposes such as smoothing, sharpening,

edge detection, and noise reduction.

6. What is the purpose of the Sobel and Laplacian operators in edge detection?

Answer: The Sobel operator detects image gradients and is commonly used to identify

edges in horizontal and vertical directions.

The Laplacian operator uses second-order derivatives to detect rapid changes in intensity.

Both operators are useful for identifying edges and fine details in an image.

Sobel generally provides directional edge information, while Laplacian detects edges

without focusing on only one direction.

7. Why are filtering operations considered an essential preprocessing step in computer vision?

Answer: Filtering operations improve image quality before further computer vision processing.

They can reduce noise, smooth unwanted variations, enhance edges, and preserve important features.

Better image quality can improve the performance of tasks such as feature extraction,

image segmentation, object detection, and object recognition.

Therefore, filtering is an important preprocessing step in computer vision.

8. Discuss the trade-off between image smoothing and edge preservation during filtering.

Answer: Image smoothing reduces noise and unwanted details, but excessive smoothing can

remove important edges and fine structures.

On the other hand, preserving edges helps maintain important object boundaries and features,

but insufficient smoothing may leave noise in the image.

Therefore, a suitable filter and kernel size must be selected to achieve a balance

between noise reduction and edge preservation.

9. Mention four real-world applications where spatial filtering techniques are widely used.

Answer: Spatial filtering is widely used in medical imaging, surveillance systems,

remote sensing, and autonomous vehicles.

In medical imaging, filtering helps reduce noise and enhance important structures.

In surveillance, it can improve image quality and enhance object boundaries.

In remote sensing and autonomous systems, filtering helps improve images for further analysis,

detection, and recognition tasks.

10. Compare spatial domain filtering with frequency domain filtering in terms of implementation and practical applications.

Answer: Spatial domain filtering directly modifies image pixels using neighboring pixel values

and convolution kernels.

It is relatively simple to implement and is commonly used for smoothing, sharpening,

noise reduction, and edge detection.

Frequency domain filtering first transforms the image into the frequency domain,

usually using the Fourier Transform, and then modifies its frequency components.

Spatial filtering is often convenient for local image operations, while frequency domain

filtering is useful for analyzing and controlling different frequency components of an image.

OUTPUT


Original Image
Grayscale
Gaussian Filter
Median Filter
Average Filter
Laplacian Filter
Sobel X
Sobel Y